# JINGYU LU

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## EDUCATION

**Peking University** 2022–2026
BSc in Intelligent Science and Technology (**Zhi Class**)
GPA: 3.46/4.00
Rank: 32/67
Selected into the prestigious **Zhi Class**, an **elite AI-oriented honors program** admitting only the **top 9.7%** of EECS students at Peking University, with advanced training in artificial intelligence and interdisciplinary innovation.

## PUBLICATIONS

**Lu, Jingyu**, Haonan Wang, Qixiang Zhang, and Xiaomeng Li. "A Cognitive Process-Inspired Architecture for Subject-Agnostic Brain Visual Decoding." *arXiv preprint arXiv:2511.02565* (2025). **Accepted at ICLR 2026.**
Haonan Wang, **Jingyu Lu**, Hongrui Li, and Xiaomeng Li. "ZEBRA: Towards Zero-Shot Cross-Subject Generalization for Universal Brain Visual Decoding." *arXiv preprint arXiv:2510.27128* (2025). **Accepted at NeurIPS 2025 (CCF-A).**

## COMPETITIONS

**VLM3D Challenge (2/50+)** 2025
Served as **project leader**. The VLM3D Challenge is an international medical AI competition co-organized by leading universities and institutions including the University of Zurich, Harvard University, Johns Hopkins University, Imperial College London, the National Institutes of Health, NVIDIA, and University Hospital Zurich. Our team **ranked 2nd** among 50+ teams from top global institutions such as UCL and Northwestern University, and later presented the results at the international MICCAI Conference.
**Chinese Chemistry Olympiad Gold Medal (19/438)** 2021
Achieved **19th place** nationwide in the Chinese Chemistry Olympiad.

## AWARDS

**Zhi Class Scholarship** 2025
**Top 3.2%** in PKU EECS, awarded to top students in the **Zhi Class**, recognizing outstanding achievements in research and academics.
**Peking University Undergraduate Freshman Scholarship** 2022
**Top 15%** in PKU, awarded to students admitted to Peking University for their outstanding academic performance among the national cohort.

## PROJECTS

### Research Projects

**CT Report Generation via Generative Models** Aug 2025 - Sep 2025
*Co-Project Lead*

- Developed CT report generation models leveraging large language models.
- Presented research outcomes at the **MICCAI** conference.

**Dynamic Brain Decoding Inspired by Cognitive Processes** Jun 2025 - Aug 2025
*Project Leader*

- Served as **project leader**, coordinating the overall research development and progress.
- Conducted detailed qualitative and quantitative experiments to validate the proposed cognitive-process-inspired decoding framework.

**Cross-Subject Brain Visual Decoding** Mar 2025 - May 2025
*Second Contributor*

- Participated in framework design discussions and implementation of brain decoding pipelines.
- Developed feature visualization and conducted qualitative analysis to interpret decoding results.

**Robot Motion and Grasping Strategies in Cluttered Environments** Jun 2024 - Dec 2024
*Second Contributor*

- Investigated AI-based planning algorithms for robotic motion in complex and cluttered scenes.
- Built simulated environments to conduct motion and grasping experiments.
- Tested and debugged real-world robot systems for experimental validation.

**Selected Course Projects**

**Enhancing Value Decomposition for Multi-Agent Cooperation via Structural Biases**
*Multi-Agent System (Final Score: 93)*

- Compare the effectiveness of different value-based MARL algorithms in handling complex cooperative environments such as the Honor of Kings Mini Environment.
- Investigate how structural modifications affect generalization, cooperation, and inter-agent interaction modeling in multi-agent reinforcement learning.

**ToM-Augemented Reinforcement Learning in Hanabi Games**
*Cognitive Reasoning (Final Score: 90)*

- Explore how incorporating Theory of Mind into reinforcement learning can enhance agents' understanding of others' beliefs and intentions in cooperative tasks.
- Evaluate whether the proposed ToM-augmented PPO framework improves multi-agent coordination and overall performance in the Hanabi environment.

## WORK EXPERIENCE OR INTERNSHIP

**Teaching Assistant** Sep 2025 -
*Class: Introduction to computing (C)*
Assisted in teaching an introductory computer science course, tutoring students in fundamental computer concepts and basic programming skills.

**UG Visiting Internship Student** Aug 2025 - Sep 2025
*Advisor: Xiaomeng Li*
Applied AI techniques to automate CT report generation and participated in a highly recognized international competition in the medical AI domain.

**Agibot, Internship** Jun 2024 - Dec 2024
*Advisor: Hao Dong*
Design of algorithms for cluttered environments, motion control for embodied intelligence, and scene construction within simulators.

**Lab Rotation, Zhi Class** Feb 2024 - Jun 2024
*Advisor: Yitao Liang, Liwei Wang, Jiahuan Zhou*
Conducted internships and research training in multiple laboratories, covering a wide range of fields including computer vision, multimodal learning, large language models (LLMs), and AI for medicine.

## EXTRACURRICULAR ACTIVITIES

**DAI Society, Yuanpei College**
**Vice President** Sep 2024 – Jun 2025
Planned and organized major academic and community events, including graduate school sharing sessions, international exchange forums, and the freshman mentor–mentee program. Represented students in Computer Science, Data Science, and AI in regular communication with faculty.

**Billiards Team, Yuanpei College**
**Captain** Sep 2023 – Jun 2024
Led the newly established Yuanpei College Billiards Team, overseeing recruitment, training, and team-building. Fostered collaboration and strong team spirit, ultimately guiding the team to win the **PKU Cup Championship**, the highest-level billiards competition at Peking University.

**Social Practice and Student Organizations Department, Yuanpei College**
**Vice President** Sep 2023 – Jun 2024
Led member recruitment and training, and organized social practice activities such as museum and aquarium visits to enrich students' academic and cultural experiences and strengthen community engagement.

## VOLUNTEER WORK

**China3DV 2025 Conference Volunteer** April 2025
Assisted with on-site coordination, attendee registration, and crowd management, ensuring efficient check-in and smooth conference operations.

## SKILLS AND INTERESTS

**Technical Skills:** Proficient in Python, C++, Git, LaTeX, MYSQL, ...
**Languages:** Mandarin, English(TOEFL:101, CET6:558)
**Interests:** Fitness, bouldering, billiards, hiking